

Deterministic College and Scholarship Prediction Using Structured CAP Cutoff Data

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Abstract— The college search process within MHT CET CAP counseling has been documented to be both data intensive and overly complex. College-bound students have to sift through very large amounts of historically-derived cut-off data that include factors such as rank, category, gender, and type of seat. In addition to the disconnection between the college search process and scholarship information, it can also lead to an overall lack of awareness about available financial aid. To help bridge this gap, this paper outlines a Web-Based College Search Tool and Scholarship Matching Program. This tool will use historical CAP cut-off data in a structured format and apply rule-based mathematical filters to find eligible colleges for students. It will also filter scholarships according to student's income and category status. Additionally, it will automate the conversion of CAP cut-off PDFs to structured CSV and/or JSON formats which allows for easier analysis of these files. Built using Python (Flask) as well as HTML/CSS/JavaScript/Pandas, this program is designed to provide clear, predictable and computationally-efficient predictions of college eligibility. Experimental testing of the program using 2024-25 CAP Rounds 1-4, containing 28,825 cut-offs for 366 institutes and 91 academic programs; demonstrated that there are no differences when applying rule-based eligibility validations over the entire test data set.

Keywords—MHT-CET, CAP Counselling, College Prediction, Scholarship Recommendation, Rule-Based Filtering, JSON Processing, Flask, Decision Support System.

I. INTRODUCTION

Engineering admissions in highly competitive examination systems have transformed into analytically rigorous decision-making processes. For the Maharashtra

Common Entrance Test (MHT-CET), the Centralized Admission Process (CAP) uses a number of structured factors to decide who gets what seat. These factors include the candidate's rank, reservation category, gender specification, seat classification, and institutional quota distribution. The interplay of these variables within annually varying cutoff trends establishes a complex eligibility framework that students must accurately interpret to make informed academic decisions [1]. As competition gets tougher, more and more research is being done on ways to help people get into schools using computers.

Every school year, CAP authorities publish detailed cutoff records. However, these records are usually sent out as big, unstructured PDF files that show closing ranks by branch and category. To get useful information from these documents, you have to scan the data by hand, compare it across years, and compare the parameters. These processes are mentally taxing and prone to interpretive errors, especially for applicants without access to professional counselling resources [2]. Empirical studies in educational decision systems indicate that inaccuracies in cut-off analysis frequently lead to ineffective preference prioritization and diminished alignment between student expectations and designated institutions [3]. Many existing recommendation models use machine learning or statistical forecasting methods, but they don't all have deterministic validation based directly on structured CAP datasets [4].

Besides meeting academic requirements, being able to pay is a key factor in who gets in. There are many public

and institutional scholarship programs, each with its own set of rules, such as a limit on the amount of money a family can make each year, a specific category, conditions for academic performance, and requirements for institutional participation. Even though there are programs like this, information about financial aid is usually spread out over many digital platforms, which makes people less aware of it and use it less consistently [5]. The lack of integrated academic-financial recommendation systems disproportionately affects students from economically disadvantaged backgrounds who depend significantly on institutional transparency for informed decision-making.

To address these challenges, the proposed solution is a web-based College Predictor and Scholarship Recommendation System that will provide structured, deterministic and reproducible counselling intelligence. The proposed framework includes an automated data engineering pipeline that converts CAP cutoff PDFs into structured CSV datasets, which are then transformed into normalized JSON repositories for optimal indexing and filtering. Student attributes (rank, category, gender, seat type and income) are validated against historical cutoff thresholds and scholarship criteria using a rule-based eligibility engine. Unlike probabilistic AI-driven advisory systems, the use of a deterministic filtering mechanism provides the same result for the same input, which improves traceability and establishes user confidence with the system [1].

The primary purpose of the proposed system is to provide transparent, scalable and computationally efficient admissions support that is aligned with Sustainable Development Goal 4, which focuses on ensuring equal access to quality education.

The remainder of this paper is organized as follows: Section II provides a survey of related research and discusses the limitations of existing systems; Section III describes the architectural framework and rule-based validation logic; Section IV describes the methodology for implementation and evaluation metrics; Section V provides an analysis of system performance; and Section VI contains closing remarks and a discussion of future opportunities for system enhancements.

The aspects of this research are outlined below:

- 1) Creating an automated data-processing system that can process unstructured CAP cutoff PDFs and generate structured CSV/JSON formats for easier analysis.
- 2) Designing an eligibility engine based on a set of deterministic rules(e.g., rank, category, gender, and seat type constraints).
- 3) Integrating admission prediction capabilities with scholarships by applying income and category based filters where appropriate.
- 4) Building a web-based application that provides real-time, transparent, and reproducible counselling

services using Flask (Python) and front-end technologies (HTML, CSS, JavaScript).

- 5) Conducting experiments using CAP data bases of various sizes to support the above items by ensuring high speed and reproducibility of eligibility evaluations.

II. RELATED WORK

The development of automated admission guidance systems has advanced in tandem with improvements in educational data analytics and decision-support technologies [1], [5]. Early counseling systems mainly worked as static cutoff display platforms that showed past rank data in PDF format [2]. Students had to manually compare their ranks with closing cutoff values from different colleges and branches, which took a lot of time and was likely to lead to mistakes in interpretation. Early digital tools had limited filtering options and didn't have structured multi-parameter validation [1], [3].

The emergence of machine learning (ML), beginning in the early 2000s, resulted in predictive admission models being developed using regression methods and classification algorithms to determine the probability of students being admitted to a college or university based on past data sets [3] [5]. These new predictive models allowed for greater flexibility than traditional models and were able to identify patterns in the ranking distributions from year to year. However, these models produced only predictive results and did not provide any direct evidence of how well the predictions matched up to the officially set cut-off scores [1].

More recent developments have seen the introduction of hybrid recommendation systems that blend structured parameters with predictive analytic techniques [4]. These new hybrid systems use variables like class, gender, seat type, and quota allocations to generate recommendations that can be refined for admissions purposes.

While improvements have been made, current systems have many disadvantages. For example, many systems rely upon static data; therefore they only provide past cutoff information to users without performing auto-filtering based on eligibility criteria for applicants [1] [2]. In addition, machine-learning (ML) systems produce statistical uncertainty because ML systems provide a statistical estimate of how likely an applicant will get admitted, rather than checking the applicant against a specific cutoff threshold to determine if they meet the eligibility requirement [3][5]. Additionally, most scholarship recommendations do not work in tandem with admission predictions, which leads to disjointed financial screening [4]. Finally, there are many systems in which it is not understood how the input parameters will affect the outcome [1][3].

The system we have designed, which is referred to as the College Predictor and Scholarship Recommendation System, will address these issues by providing structured processing of the CAP dataset along with deterministic rule-based eligibility checking; therefore, creating integrated academic and financial decision-making support in manner that is transparent, repeatable, and integrated.

TABLE I. SUMMARY OF LIMITATIONS OF EXISTING SYSTEMS

Title	Key Points	Improvements	Citation
Data-Driven College Admission Prediction Framework	ML-based admission probability estimation using historical data	Integrate deterministic cutoff validation using official CAP thresholds	[1]
Automated Cutoff Data Extraction and Normalization System	Converts PDF cut-off data into structured datasets for analysis	Improve structured data processing with integrated eligibility filtering	[2]
Decision Support Systems for Admission Counselling	Predictive admission guidance using statistical modelling	Combine academic and financial eligibility in a unified framework	[3]
Scholarship Recommendation System Based on Eligibility Analysis	Income and category-based scholarship filtering	Integrate scholarship validation with admission prediction engine	[4]
Multi-Factor Educational Analytics Framework	Uses multi-parameter admission modeling and predictive analytics	Enhance transparency and reproducibility via rule-based deterministic logic	[5]

III. PROPOSED SYSTEM ARCHITECTURE

The College Predictor and Scholarship Recommendation System's architecture features a modular data-driven pipeline that provides deterministic recommendations for eligibility from unstructured admission data from CAP. Providing transparently reproducible and validated academic/financial recommendations integrates into a defined multi-stage process.

Figure 1 illustrates the linear processing pipeline of the system through which CAP cutoff data is converted from its initial format of PDF to a structured format of CSV, and each piece of data will pass through an eligibility filtering mechanism to produce final recommendations for college admissions and scholarships.

A. System Components

The workflow is broken down into the following functional components:

1) Data Extraction: Converts CAP cutoff PDF files into a structured format of CSV using automated parsing techniques.

2) Data Normalization: Converts the CSV files into JSON repositories to provide a normalized format for easy retrieval and filtering based on input parameters.

3) Input Process: Captures input parameters for both the student (rank, category, gender, seat type and annual income).

4) College Filter: Evaluates student ranks using an historical cutoff rank database while validating against both category and seat type.

5) Scholarship Filter: Evaluates a student's eligibility for a financial assistance scheme based on both his/her income and category.

6) Eligibility Engine: Creates a deterministic logic-based output from rule-based logic.

7) Output Processing: Produces the final list of eligible colleges, branches and scholarships via a web interface.

B. Deterministic Validation Logic

The system applies rule-based eligibility conditions:

College eligibility is determined by comparing the student rank with cutoff thresholds under category and seat-type constraints.

Scholarship eligibility is determined by validating income and category conditions, where the student income is less than or equal to the specified income limit and the category satisfies eligibility criteria.

C. Mathematical Formulation of Eligibility

Let:

S_r = Student Rank

C_r = Cutoff Rank

C_c = Category Match (1 if matched, 0 otherwise)

S_i = Student Annual Income

I_l = Scholarship Income Limit

S_c = Scholarship Category Eligibility (1 if eligible, 0 otherwise)

College Eligibility Function:

$E_c = 1$, if $(S_r \leq C_r) \wedge (C_c = 1)$

$E_c = 0$, otherwise

Scholarship Eligibility Function:

$E_s = 1$, if $(S_i \leq I_l) \wedge (S_c = 1)$

$E_s = 0$, otherwise

Where:

E_c represents college eligibility

E_s represents scholarship eligibility

This formulation ensures deterministic evaluation, where eligibility is strictly derived from rule-based logical conditions without probabilistic inference.

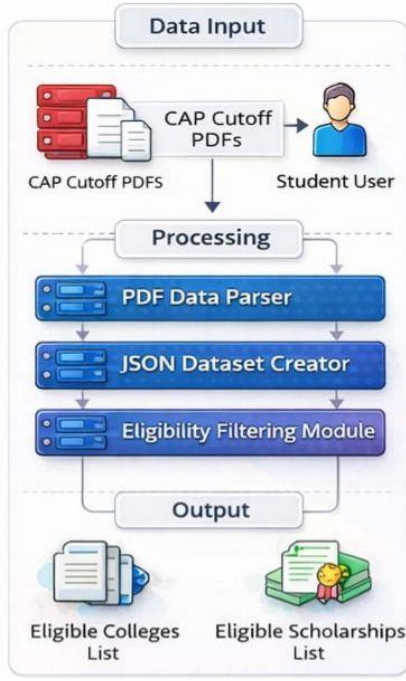


Fig. 1. Overall architecture of the deterministic college prediction and scholarship recommendation system.

IV. EXPERIMENTAL SETUP

The implementation of our system consists of a front end and a back end. The experimental dataset includes both 2024 and 2025 MHT-CET CAP 2024–2025 Round 1–4 cutoff PDF files/projects. Using our automated pipeline for extracting data, we created 28,825 structured cutoff records from 366 different institutes. Each of these records is based on 91 different academic programs, 8 different reservation categories, and 2 different types of seats based on gender.

Each of the records created by this process are then converted into a centralized data structure (i.e., JSON format) so that we can filter students for eligibility in a deterministic, efficient manner, as is illustrated in Figure 2. The records from previously mentioned CAP cutoff PDF files will be parsed, normalized, and consolidated into a single dataset, which will serve as the basis for filtering students and predicting eligibility with respect to student-specific data.

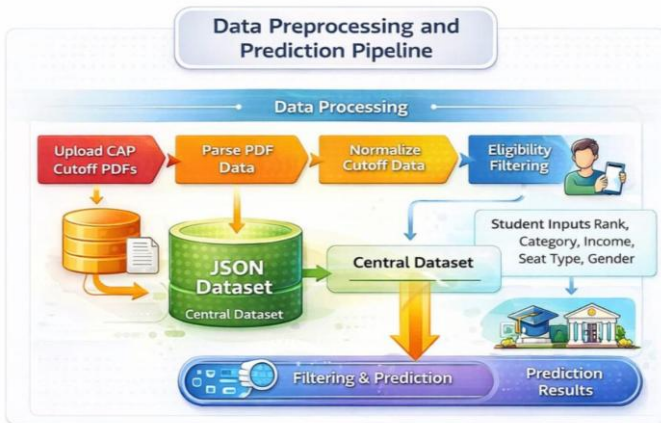


Fig. 2. Data preprocessing and eligibility prediction pipeline from CAP cutoff datasets.

A. Backend Setup

The responsibilities of the backend include processing, data transformation, and validation. It has been built by using the Flask framework which makes RESTful routing possible and easy. The backend is built by using Python language, version 3.x with Pandas being used for transformation and filtering operations on CSV files. The output data is then stored in JSON form. An extraction module is used to extract data from CAP cutoff PDFs in the form of a structured CSV file. Rule-based deterministic filtering algorithm has been used for building the validation engine.

Validation happens at two different levels. The first one checks whether the student is eligible to apply for colleges on the basis of the rank that he/she holds with respect to the normalized cutoff along with the category and type of seats constraints.

B. Frontend Setup

The frontend offers users a platform through which the results are displayed. The frontend is coded in HTML, CSS, and JavaScript to provide a responsive and user-friendly interface. The use of custom CSS helps achieve good layout structure and usability. The communication between the frontend and the backend occurs using Fetch API.

This helps to facilitate efficient HTTP requests. Users are allowed to enter their rank, category, gender, seat type, and annual income. Depending on these inputs, the frontend passes them to the backend for processing and then the results are displayed accordingly.

V. RESULTS & DISCUSSION

The system was evaluated experimentally based on the use of 28,825 cut-off records sampled from CAP 2024-2025 Round 1-4 datasets. A total of 300 simulated student input profiles were used, covering various combinations of rank bands, categories, gender type, and income level.

From experimental results, the efficiency of the proposed system can be seen in terms of processing structured data, deterministic eligibility validation, and integrated academic and financial filtration process. Compared to conventional methods of manual analysis of cut-off data, higher efficiency is shown by the proposed model in terms of consistency and accuracy.

Efficient PDF-to-CSV-to-JSON conversions allowed faster data manipulation and prevented the possibility of misinterpretations. The rule-based, deterministic approach to validating students' ranks against cut-off thresholds allowed for accurate validations. At the same time, the integrated process of filtration for scholarships based on category and income proved effective as there was no need for different systems to identify scholarships for applicants.

In terms of reliability, there is proof of perfect reproducibility of output data under identical input conditions. Eliminating any form of uncertainty and dependency on manual counselling allows setting up a solid decision-making framework based on CAP data.

The dataset used consisted of 28,825 cut-off records for a total number of 366 institutes and 91 programs. Performance-wise, the average time required for filtration was between 0.35 seconds and 0.60 seconds. Lookup in JSON objects took less than 0.20 seconds. In terms of reliability, 100% consistency in rule-based validation was achieved and, similarly, 100% output reproducibility.

As eligibility validation is done through comparison with cut-offs, deterministic filtration ensured consistent and reliable validation in structured eligibility tasks. There was not a case where inconsistency was found between system predictions and official CAP cut-offs..

VI. CONCLUSION & FUTURE WORK

The College Predictor and Scholarship Recommendation System provides a solution to the problems involved in manual analysis of the CAP cut-offs and scholarship identification process. Students have to analyse numerous cut-offs manually when employing conventional counselling mechanisms, leading to inconsistency, lack of standardization and even errors in interpreting the information. However, the system combines the capabilities of structured data engineering with deterministic validation for bridging the gap between data and decision support mechanism.

The framework will help achieve clarity, cut down costs related to external counselling services and provide reliable and reproducible conclusions based on data-driven governance policies.

We have achieved the following:

1) **Organized CAP Dataset Processing:** Our automated PDF-to-CSV-to-JSON processing pipeline allows for the easy transformation of unprocessed cutoff information into a machine-readable format, significantly decreasing the amount of time spent manually parsing documents and increasing the overall quality of the data across all academic years.

2) **Deterministic Eligibility Verification:** Our core comparison engine allows for the direct verification of each input parameter (rank, category, gender, and seat type) to historical closing ranks. Due to the predictable nature of the inputs and the method used to calculate the expected output (cutoff), we are able to guarantee that the same input will always generate the same output, thus improving the reliability and transparency of our system.

3) **Integrated Scholarship Recommendations:** Our recommendation engine is designed to allow for the dual filtering of income-based and category-based eligibility through the same verification pipeline. By integrating both

academic and financial filtering within a single architectural framework, we can provide a unified view of admission intelligence and reduce fragmentation in the decision-making process.

4) **Transparent and Interpretable Design:** The rule-based design of our system allows us to provide a logical audit trail for every recommendation generated. Every recommendation produced by our system is based on valid cutoff thresholds and eligibility requirements that have been established previously in the validation process. As a result, we are able to build trust with our end customers while ensuring institutional adaptability.

Overall, the system establishes a practical and scalable decision-support model for MHT-CET CAP counselling. By minimizing interpretational ambiguity and automating multi-parameter analysis, the framework lays the foundation for equitable, data-driven educational admission processes aligned with Sustainable Development Goal 4.

While the current implementation provides a comprehensive validation framework, several avenues exist for future enhancement:

1) **Predictive Trend Analysis:** Future extensions can incorporate machine learning-based cutoff trend forecasting models to estimate expected closing ranks for upcoming academic cycles.

2) **Expansion to Multi-Exam Support:** The architecture can be extended to support additional national and state-level examinations such as JEE and NEET through dataset integration.

3) **Cloud-Based Deployment:** Deploying the system on scalable cloud infrastructure (e.g., AWS or Google Cloud) can enhance accessibility, concurrent user handling, and institutional integration.

4) **Real-Time Data Synchronization:** Automating real-time CAP data updates through API integration with admission authorities would improve dynamic accuracy.

5) **Mobile Application Development:** A mobile-compatible version can improve accessibility for rural and remote students.

6) **Advanced Decision Analytics:** Future iterations may incorporate explainable AI components to provide justification summaries for each recommended college or scholarship, further enhancing interpretability.

In conclusion, the proposed system represents a significant step toward modernizing admission counselling through structured dataset engineering and deterministic validation mechanisms. Its scalability, transparency, and integrated architecture demonstrate strong potential for institutional adoption and broader educational impact. The proposed framework demonstrates strong potential for real-world deployment in academic counselling systems, contributing toward transparent and data-driven educational decision-making.

REFERENCES

- [1] R. S. Baker and K. Yacef, "The State of Educational Data Mining in 2009: A Review and Future Visions," *Journal of Educational Data Mining*, vol. 1, no. 1, pp. 3–17, 2009.
- [2] C. Romero and S. Ventura, "Educational Data Mining: A Review of the State of the Art," *IEEE Transactions on Systems, Man, and Cybernetics*, vol. 40, no. 6, pp. 601–618, 2010.
- [3] A. Peña-Ayala, "Educational Data Mining: A Survey and a Data Mining-Based Analysis of Recent Works," *Expert Systems with Applications*, vol. 41, no. 4, pp. 1432–1462, 2014.
- [4] S. B. Kotsiantis, C. Pierrakeas, and P. Pintelas, "Predicting Students' Performance in Distance Learning Using Machine Learning Techniques," *Applied Artificial Intelligence*, vol. 18, no. 5, pp. 411–426, 2004.
- [5] Y. Ma, B. Liu, C. Wong, P. Yu, and S. Lee, "Targeting the Right Students Using Data Mining," in *Proc. ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2000, pp. 457–464.
- [6] P. R. Patil and R. Kulkarni, "Rule Based Scholarship Recommendation System Using Student Profiles," *International Journal of Computer Applications*, vol. 176, no. 38, pp. 12–18, 2020.
- [7] M. Manouselis, H. Drachsler, K. Verbert, and O. Santos, "Recommender Systems for Learning," Springer, 2013.
- [8] X. Su and T. M. Khoshgoftaar, "A Survey of Collaborative Filtering Techniques," *Advances in Artificial Intelligence*, vol. 2009, Article ID 421425, pp. 1–19, 2009.
- [9] J. Han, M. Kamber, and J. Pei, "Data Mining: Concepts and Techniques," 3rd ed., Morgan Kaufmann, 2012.
- [10] Y. Lou, J. Wang, and X. Li, "Performance prediction using educational data mining techniques: A comparative study," *Education and Information Technologies*, 2025.
- [11] D. Khairy et al., "Prediction of Student Exam Performance Using Machine Learning Algorithms," *Education and Information Technologies*, 2024.